

AST 4762/5765: Astronomical Data Analysis

Installing WSL for Windows & running VS Code.

CAREFUL: If you cut-and-paste from a PDF document, likely all non-alphanumeric characters, including commas, apostrophes, quotation marks, underscores, dashes, hyphens, tabs, symbols like ~, and so on will look right but work wrong. This is because typesetting programs use prettier versions of these characters, and the shell doesn't understand anything but simple ASCII characters. Either retype these symbols or just type the whole line by hand.

It is important to log what you do when you're doing system management! Especially log anything that deviates from these instructions.

A. The automated way to install WSL [with thanks to David Wright]:

1. Getting Windows Ready

In order to get a graphical install of Linux working in the Windows Subsystem for Linux (WSL), we will need version 2. WSL2 is only available in Windows 10, Version 2004, Build 19041 or higher. To find out which version of Windows your device is running, press the **Windows logo key + R**, type **winver** in the **Open** box, and then select **OK**. If you do not meet the requirements, please update your machine before proceeding.

2. Installing WSL2

Open PowerShell as Administrator and run (type all of the commands in this document manually, do not copy paste):

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

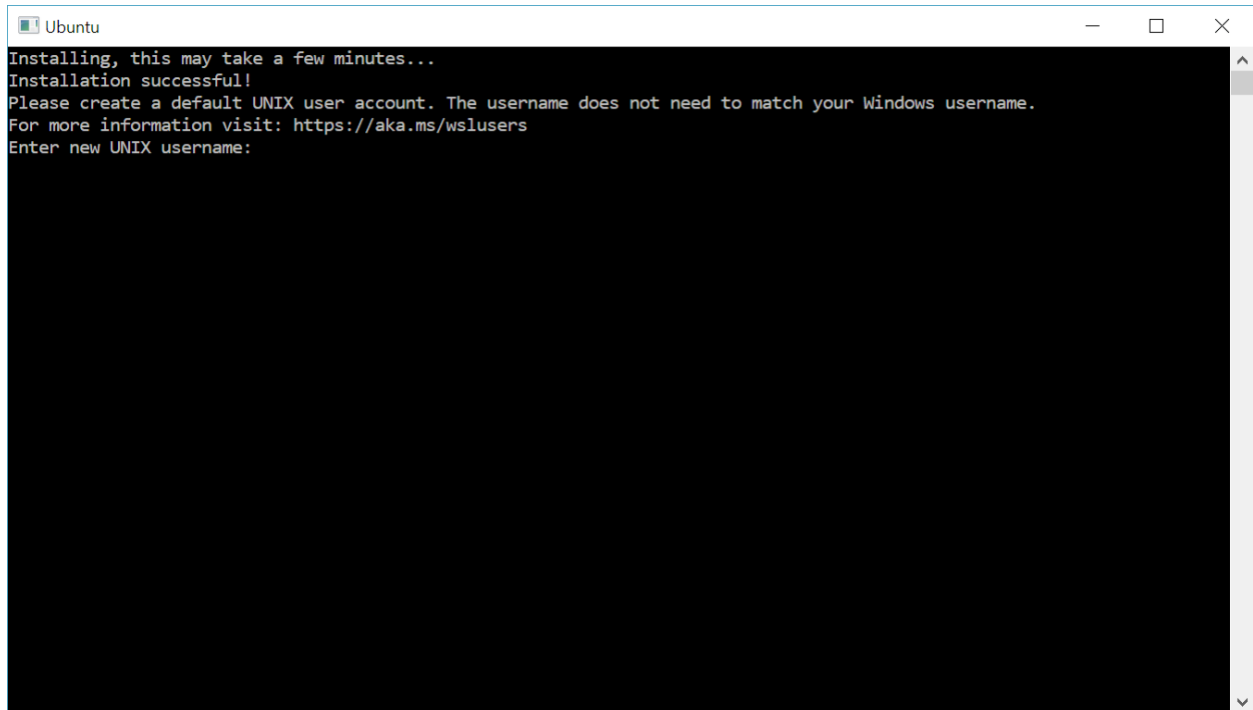
Now, restart your machine. Once again open PowerShell as Administrator and run the following command

```
wsl --set-default-version 2
```

You might see this message after running that command: **WSL 2 requires an update to its kernel component**. If you do, visit <https://aka.ms/wsl2kernel> and install the latest WSL2 Linux kernel. Once you have the kernel installed, run the above command again, and it should complete successfully without showing the message.

3. Installing Ubuntu 20.04

Open the Microsoft Store and download Ubuntu 20.04 LTS. Once it is downloaded, launch it like you would a normal application. You will see something that looks like the following picture:



Create a username and password.

4. Getting a Desktop Environment Working

We will need to transfer a file from our Windows file system to our Linux file system. The Windows C:\ drive is mounted by default at `/mnt/c/`. Go to that directory with the following command,

```
cd /mnt/c
```

Locate the `wsl-install.sh` script from the WebCourses files. Copy it to our Linux home directory with

```
cp <path to wsl-install.sh> ~/
```

Run this script:

```
cd ~
```

```
sudo bash wsl-install.sh
```

You will be prompted to choose a display manager, choose **gdm3**. This script will take a while to run. When the script finishes, close your WSL window. Then, relaunch it.

5. Starting the Desktop Environment

Now that the script has finished, start the service with

```
session start
```

Open the built-in Windows 10 remote desktop program. Enter **localhost:3389** in the **Computer** box, then hit **Connect**. Click **yes** if you are asked if you want to connect. Log in with your Linux username and password.

6. Tweaking the XFCE Desktop

The desktop environment (XFCE in our case) attempts to put the display to sleep after 10 min. This causes our environment to become unresponsive. To fix this, head into the lock screen settings in XFCE. You can find this by clicking the magnifying glass icon in your dock and searching for "Light Locker". Change your settings to reflect this picture, then apply ("Lock on suspend" will remain on, just make sure to turn off "Enable light-locker", "Blank screen after" and "Switch off display after" to "Never").



Congratulations! You should be able to follow the linux software install guide from now on.

B. The manual (older) way to install WSL

If you have an older Windows build you can try the manual installation from the [Win10-software.pdf](#) or the Windows link here: <https://learn.microsoft.com/en-us/windows/wsl/install-manual>

C. Updating your Ubuntu packages

Any time you get a new operating system, the first thing you should do is update the software to get any security fixes or improvements since the version you installed. Run the following:

```
sudo apt update  
sudo apt upgrade
```

Here, sudo (Substitute the administrator's Userid and DO something) will make the command run as an administrator ("root" in Unix-speak). It will ask for your password. The first command updates the database of packages available. The second actually updates the packages. You will have additional prompts to confirm doing the update. Press enter to these. It will take a few minutes to update.

Using VS Code

You can use VSC to work directly on your Unix section. For instruction on how to access it see here: <https://code.visualstudio.com/docs/remote/wsl>